

Nathan Zhao

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Education

Georgia Institute of Technology <i>Bachelor of Science in Electrical Engineering</i>	Atlanta, Georgia
The Pembroke Hill School <i>High School Diploma, Cum Laude</i>	Kansas City, Missouri
• Relevant Coursework: ACT: 36; Multivariable Calculus, Differential Equations, Linear Algebra	

Selected Projects

Design and Development of a Bimanual Mobile Manipulator for Assisted Living • Onshape, C++, PlatformIO, $\LaTeX$	2024 - Present
LeSCARA: Low-Cost Open-Source SCARA Platform for Embodied AI Research • Tools Used: Onshape, C++, Python, Pico SDK	2026 - Present
Continuous Finger Joint Angle Estimation Using Fiducial Markers and Real-Time Pose Disambiguation • Python, OpenCV, SolidWorks, C++, $\LaTeX$	2025 - 2026
mmHome: Development of a Multimodal IoT Network for Contactless Vital Sign Health Monitoring and Fall Detection • Python, scikit-learn, C++, PlatformIO	2025 - Present

Work Experience

Research and Development Intern @ MRIGlobal • Designed and fabricated custom chemically coated AGS02MA gas sensor array and STM32 PCB (Altium), enabling real-time detection and classification of chemical warfare agents • Developed embedded C++ firmware and Python GUI tools for real-time data acquisition, preprocessing, and TensorFlow Lite CNN LSTM model deployment on-device	2025
Research Assistant @ University of Missouri-Kansas City, Multiplatform Interactive Robotics Lab (mirolab) • Worked under Professor Hee Sup Shin on vision-based finger goniometry with FDM-printed ArUco wearable • Developed Python programs for Cubature Kalman fusion and filtering of MediaPipe hand landmark detection with 6DOF fiducial marker pose estimation using two cameras	2025
Research Intern @ University of Pittsburgh, Accessible SmartTech Lab • Worked under Professor Dan Ding with ROS Noetic in Ubuntu 20.04 on a Hello Robot Stretch RE1 • Programmed autonomous functionality using ArUco marker detection and 2D LiDAR SLAM with RViz	2024
Intern @ WeCode KC • Developed video game interactive demo to teach young students programming concepts using C# and Unity	2022

Relevant Skills

Programming: C++, Python, C#, Java, MATLAB, $\LaTeX$
Robotics/ML: PyTorch, OpenCV, ROS, Arduino, scikit-learn, Unity
Computer Aided Design: SolidWorks, Onshape, Altium Designer, KiCAD
Languages: English, Chinese (Mandarin), Spanish